

Supplementary material for “Optimal multi-stage designs for randomised clinical trials with continuous outcomes”

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1 Supplementary methods

1.1 Simulated annealing

For two-stage designs the number of parameters is low enough that a grid-search procedure can be used to find the δ -minimax design. As the number of stages goes up the number of parameters increases and the amount of computation needed to find the properties of a particular design increases. This makes a grid-search technique infeasible to use. An alternative method to search over a large space is to use a stochastic search technique, an example of which is simulated annealing.

Simulated annealing is described well by Lin and Geyer[1] for the context of semiparametric regression. An application to clinical design was proposed by Banerjee and Tsiatis[2] for finding the parameters used in the dynamic programming method of Barber and Jennison[3]. The application to finding the δ -minimax design is different, since we apply it to find the group-size and stopping boundaries directly.

Each iteration in the simulated annealing process consists of two steps. Firstly a new candidate design is generated. Secondly the process ‘accepts’ or ‘rejects’ the candidate design depending on the function values of the current design and the candidate design. If accepted the candidate design becomes the current design and the process repeats. If rejected, the current design remains and the process repeats. Both steps depend on the current ‘temperature’ of the system, which reduces as the number of candidate designs generated increases.

The first thing needed to apply simulated annealing is a function to be minimised. In this case the quantity of interest is the maximum expected sample size. To restrict the search to designs which have the correct significance level and power (henceforth referred to as ‘feasible’ designs), we add a penalty function which penalises designs with higher than required significance level or lower than required power. It is possible to completely disallow infeasible designs but in order to find the δ -minimax design it may be required to pass through designs which are infeasible. The function to be minimised has the following form:

$$f(d) = \mathbb{E}(N|\tilde{\delta}) + N_{\text{Single}} \left(\frac{(\alpha' - \alpha)}{\alpha} I\{\alpha' > \alpha\} + \frac{(\beta' - \beta)}{\beta} I\{\beta' > \beta\} \right), \quad (1)$$

where d is the design; $\mathbb{E}(N|\tilde{\delta})$ is the maximum expected sample size of the design; N_{Single} is the sample size required for the single-stage design; α and β are the required type I and II error probabilities; α' and β' are the actual type I and II error probabilities of d .

For generating new candidate designs, one of the stages is picked at random. The group-size and stopping boundaries corresponding to that stage are perturbed by adding independent normally distributed random variables with mean 0 and standard deviation θ (where a different value of θ is used for the group-size). If the resulting design is valid (i.e. the efficacy boundary is higher than the futility boundary), this design becomes the candidate design. Let the current design be d and the candidate design be d' . If $f(d') \leq f(d)$, d' becomes the current design. If $f(d') > f(d)$, then d' becomes the current design with probability:

$$\exp \left(-\frac{f(d') - f(d)}{t} \right), \quad (2)$$

where t is the current temperature of the system. After each candidate design generation the temperature is reduced, with the new temperature set to $\rho_1 t$ for some constant $\rho_1 \in (0, 1)$. If t is high then the process is more likely to move to worse designs. Moving to a worse design may be required in order to find the true δ -minimax design instead of a local optimum. In addition to the temperature reducing, the standard deviation θ , used to generate new designs is multiplied by a factor $\rho_2 \in (0, 1)$. This reduces the difference between candidate designs and the current design, meaning that later in the process only designs which have similar parameters to the current best one are searched.

There is no accepted best way of selecting the starting values for θ and t . For our purposes we start with $\theta = 3$ for parameters corresponding to stopping boundaries and $\frac{N_{\text{Single}}}{5}$ for the group-size. An initial value for t is taken as 100. The values of ρ_1 and ρ_2 used are such that after 10000 candidate designs are generated, t and σ are 0.005.

For the starting point of the process we use the triangular design. It is immediate to find and is generally not too far from the δ -minimax design, so seems like a good point to start the process at. To improve the reliability of the process only designs where the futility bounds increase and the efficacy bound decrease from stage to stage are permitted.

We made a few tweaks to the method in order to increase the reliability. Firstly, if 25 designs are generated without having found a new minimum, the current design is reset to the best design found so far. This prevents the process from exploring too far away from the current minimum. Secondly, after 10000 designs are generated the process restarts at the initial values for t and σ . This allows the process to start exploring again from a good design and should increase the chances of finding the best design. The minimum number of restarts is set at the beginning of the process to be 5. If subsequent restarts fail to improve the current design, the process ends. Thirdly, group-size is initially not constrained to be an integer. This improves the properties of the process but means that the final design produced has a non-integer group-size.

To overcome this, the group-size is rounded down to the nearest integer, and another simulated annealing run is started with just the futility and efficacy boundaries being changed. This produces a feasible design with integer group-size.

We have not systematically investigated the above parameters. They appear to work well in practice, but there are almost surely improvements that can be made. In the future a more thorough investigation of which values should be used for the process is planned.

2 Supplementary results

2.1 $(\alpha, \beta) = (0.05, 0.2)$

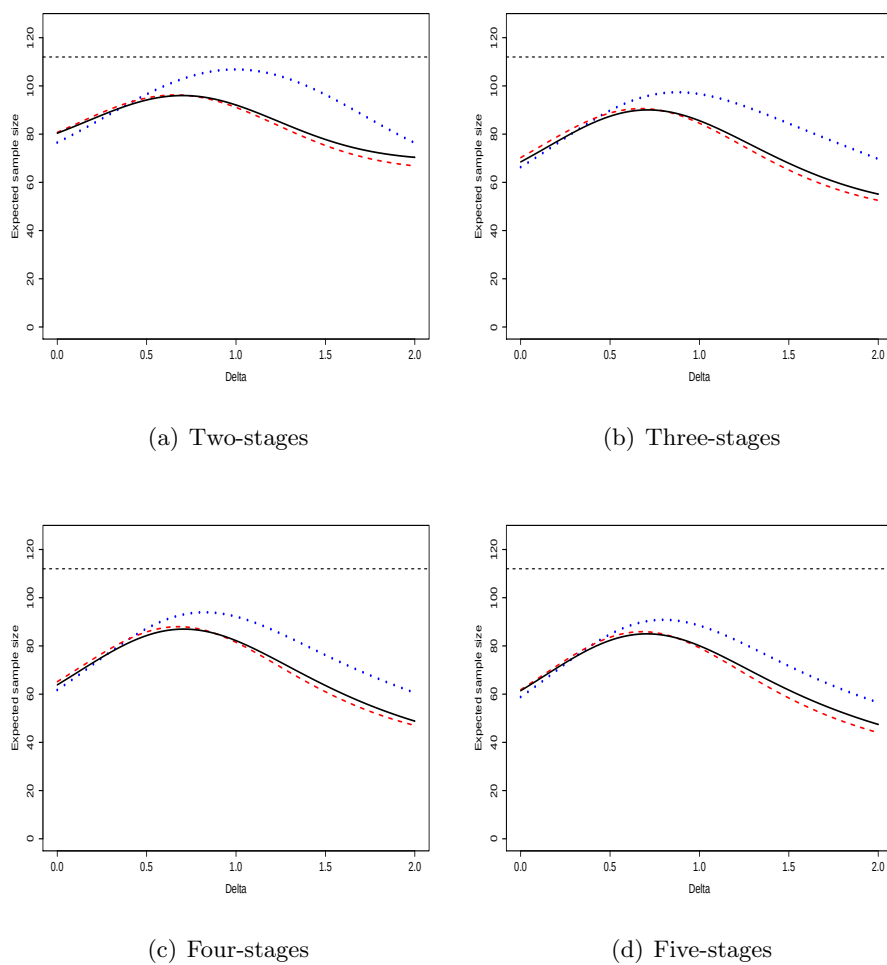


Figure 1: Plot of expected sample sizes of the null-optimal design (dotted), the CRD-optimal design (dashed), and the δ -minimax design (solid), $(\alpha, \beta) = (0.05, 0.2)$. The horizontal dashed line represents sample size of single-stage design.

2.2 $(\alpha, \beta) = (0.1, 0.1)$

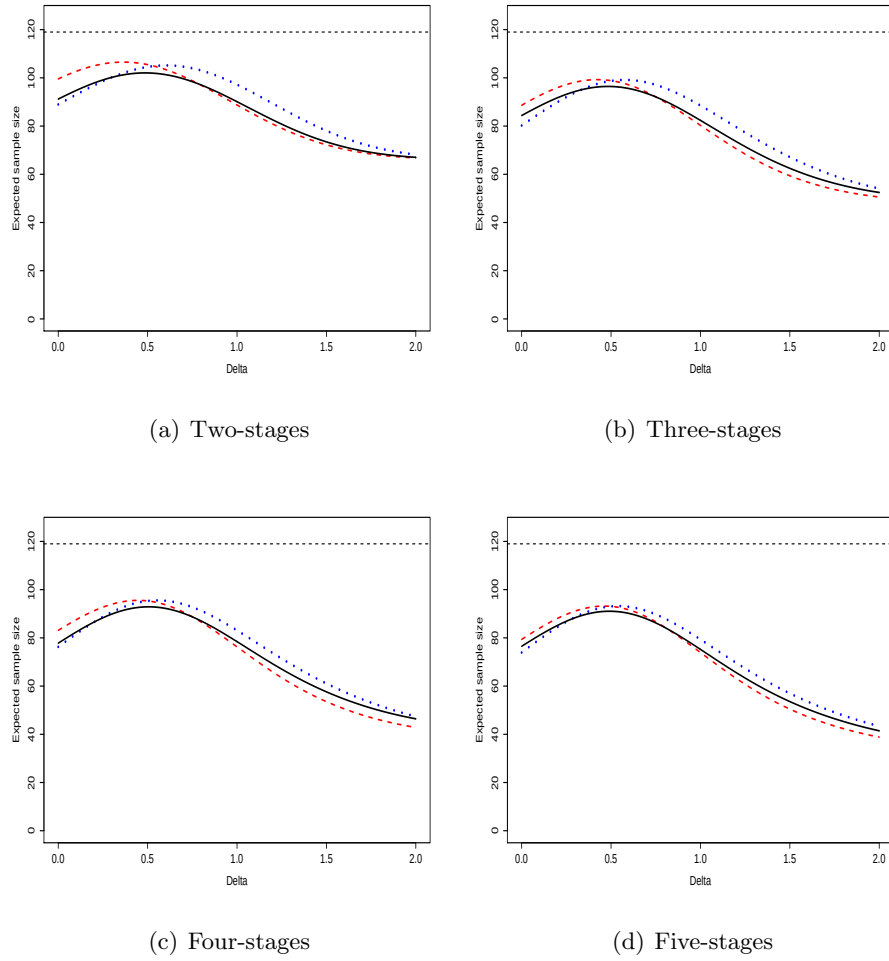


Figure 2: Plot of expected sample sizes of the null-optimal design (dotted), the CRD-optimal design (dashed), and the δ -minimax design (solid), $(\alpha, \beta) = (0.1, 0.1)$. The horizontal dashed line represents sample size of single-stage design.

2.3 Different response variances

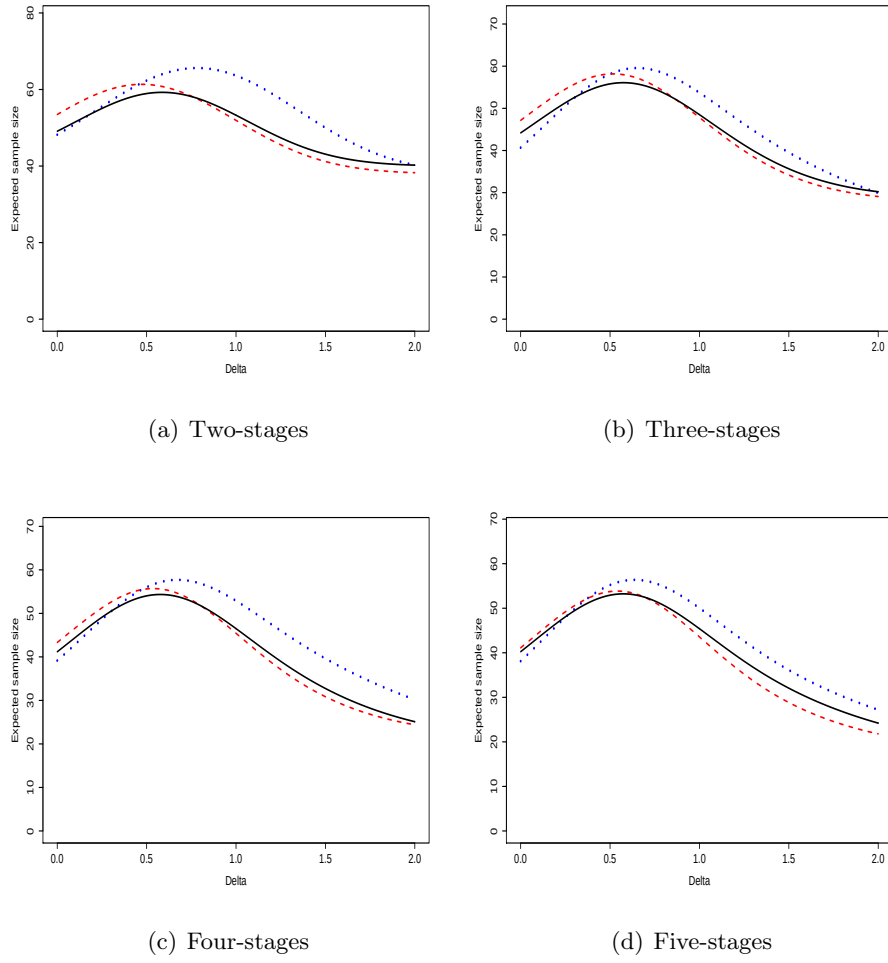
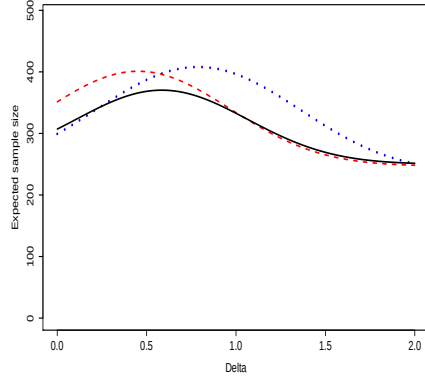
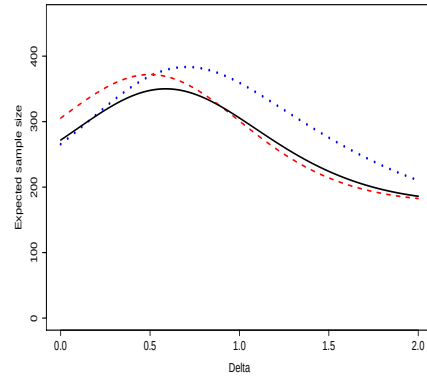


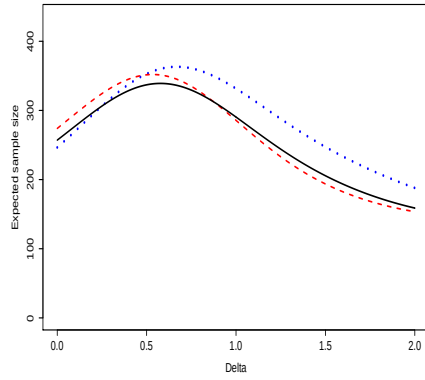
Figure 3: Plot of expected sample sizes of the null-optimal design (dotted), the CRD-optimal design (dashed), and the δ -minimax design (solid), $(\alpha, \beta) = (0.05, 0.1)$, $\frac{\sigma}{\delta_1 - \delta_0} = 2$



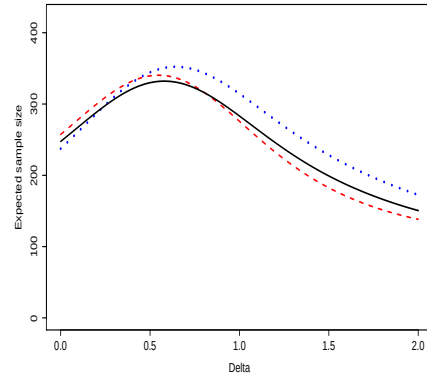
(a) Two-stages



(b) Three-stages



(c) Four-stages



(d) Five-stages

Figure 4: Plot of expected sample sizes of the null-optimal design (dotted), the CRD-optimal design (dashed), and the δ -minimax design (solid), $(\alpha, \beta) = (0.05, 0.1)$, $\frac{\sigma}{\delta_1 - \delta_0} = 5$

References

- [1] D.Y. Lin and C.J. Geyer. Computational methods for semiparametric linear regression with censored data. *Journal of Computational and Graphical Statistics*, 1:77–90, 1992.
- [2] A. Banerjee and A.A. Tsiatis. Adaptive two-stage designs in phase II clinical trials. *Statistics in Medicine*, 25:3382–3395, 2006.
- [3] S. Barber and C. Jennison. Optimal asymmetric one-sided group sequential tests. *Biometrika*, 89:49–60, 2002.